

Roll No.....

MID SEMESTER EXAMINATION March - 2025

Name of the Program: B. Tech (EE)
Name of the Course: High Voltage Engineering

Semester: 6th
Course Code: TEE 608

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Write down a short note on the thermal breakdown for solid material with suitable equations. What will the breakdown strength of air be for small gaps (1mm) and large gaps (20cm) under uniform field conditions and standard atmospheric conditions?	
OR		
(b)	Define Townsend's second ionization coefficients. How is the condition for the breakdown obtained in a Townsend discharge?	
Q2	(10 marks)	CO1
(a)	A ten stage Cockraft-Walton circuit has all capacitors of 0.06 μF . The secondary voltage of the supply transformer is 100 kV at a frequency of 150 Hz. If the load current is 1 mA, determine (i)voltage regulation (ii) the ripple (iii) the optimum number of stages for maximum output voltage (iv)the maximum output voltage.	
OR		
(b)	A 100 kVA, 400V/250kV testing transformer has 8% leakage reactance and 2% resistance on 100 kVA base. A cable has to be tested at 500kV using the above transformer as a resonant transformer at 50 Hz. If the charging current of the cable at 500kV is 0.4A, find the series inductance required. Assume 2% resistance for the inductor to be used and the connecting leads. Neglect dielectric loss of the cable. What will be the input voltage to the transformer?	
Q3	(10 marks)	CO2
(a)	What are 'Treeing' and 'Tracking'? Explain clearly the two processes in solid dielectrics.	
OR		
(b)	State and explain Paschen's law. How do you account for the minimum voltage for breakdown under a given pd condition?	
Q4	(10 marks)	CO2
(a)	Explain clearly the suspended particle mechanism of liquid breakdown with figures and suitable equations.	
OR		
(b)	Explain briefly various theories of breakdown in liquid dielectrics. What is the difference between Pure and Commercial liquids?	

Q5	(10 marks)	CO2
(a)	Describe with a neat diagram the principle of operation, application, and limitations of the Van de Graf generator.	
OR		
(b)	With a neat sketch, explain the principle of operation of a Cascaded Transformer Connection for high-voltage AC generation. What are its merits?	